Supplementary Appendix: Generated Benchmark Result Tables

The tables below are generated automatically from the recorded runs in the repository (https://github.com/tuandung222/LuatCheck-D2); a script replaces internal run identifiers with plain labels while keeping every cell value unchanged. **A.** Controller scaling comparison (Bonsai 2B vs Bonsai 4B) across both benchmarks: strict success with paired differences, then behaviour match, Recall@5, answer rate and execution-error rate. **B.** Per-benchmark result tables of the two runs with the Bonsai 4B controller.

Benchmark	Configuration	Bonsai 2B Succ. (%)	Bonsai 4B Succ. (%)	Controller effect 4B−2B (pp)
HCMUT regulations	no verifier	31.3 [22–41]	70.0 [61–79]	+38.7 [+27.3,+50.0]*
	verifier	24.0 [15–33]	64.0 [53–75]	+40.0 [+28.7,+51.3]*
	verifier + reranker	28.7 [20–37]	60.7 [50–71]	+32.0 [+21.3,+42.0]*
Labour law	no verifier	31.3 [22–41]	62.0 [52–72]	+30.7 [+20.7,+40.7]*
	verifier	28.0 [19–38]	52.7 [43–63]	+24.7 [+12.0,+37.3]*
	verifier + reranker	28.0 [19–37]	56.7 [47–67]	+28.7 [+18.0,+39.3]*

Table 1: Strict success across controller sizes (Bonsai 2B vs Bonsai 4B) on both benchmarks (LLM-judge estimate, **not human labels**; lower bound over all planned trials, errors and trials the judge could not rate count as failures; seeds averaged per question). Controller effect = 4B minus 2B (paired bootstrap differences over questions, 5000 draws). *: the 95% interval excludes 0 (descriptive, no multiple-comparison correction). Same questions in every column.

Benchmark	Configuration	Behaviour match (2B / 4B, %)	Δ controller (4B−2B)	Recall@5 (2B / 4B)	Δ controller (4B−2B)	Answer rate (2B / 4B, %)	Δ controller (4B−2B)	Execution-error rate (2B / 4B, %)	Δ controller (4B−2B)
HCMUT regulations	no verifier	58.0 / 88.0	+30.0*	0.856 / 0.989	+0.133*	55.3 / 80.7	+25.3*	0.0 / 2.0	+2.0
	verifier	45.3 / 82.7	+37.3*	0.859 / 0.989	+0.130*	41.3 / 72.7	+31.3*	0.0 / 2.0	+2.0
	verifier + reranker	46.7 / 80.0	+33.3*	0.844 / 0.978	+0.133*	42.0 / 72.7	+30.7*	0.7 / 2.0	+1.3
Labour law	no verifier	78.7 / 94.0	+15.3*	0.967 / 0.989	+0.022	74.0 / 88.0	+14.0*	0.0 / 0.0	+0.0
	verifier	63.3 / 81.3	+18.0*	0.989 / 0.989	+0.000	56.0 / 74.0	+18.0*	0.0 / 2.7	+2.7*
	verifier + reranker	65.3 / 79.3	+14.0*	0.993 / 0.993	+0.000	57.3 / 72.0	+14.7*	0.7 / 2.7	+2.0

Table 2: Metrics that need no judge across controller sizes (Bonsai 2B vs Bonsai 4B): behaviour match (expected answer → answered, expected refusal → refused), Recall@5 of gold documents in the agent's results (answerable questions; a retrieval measure on document ids, not answer correctness), answer rate and execution-error rate; percentages except Recall@5. Each cell lists 2B / 4B; Δ lists the controller effect (4B minus 2B) in percentage points (Recall@5 absolute). *: the paired 95% bootstrap interval excludes 0 (descriptive).

Benchmark	Configuration	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
HCMUT regulations	no verifier	150	70.0 [61–79]†4	84.2	89.8	0.989	0.953	0.844	0.807	0.820
	verifier	150	64.0 [53–75]†6	81.2	90.3	0.989	0.952	0.726	0.693	0.704
	verifier + reranker	150	60.7 [50–71]†10	77.1	89.3	0.978	0.937	0.733	0.704	0.714

Table 3: Results with the Bonsai 4B controller on the HCMUT regulations benchmark, three configurations (450 trials; random seeds 42, 123, 2026). *Succ.* is strict success (appropriate behaviour, correctness and completeness both full, every claim supported): a lower bound over all trials, with a 95% bootstrap interval over questions; execution errors count as failures, and †k marks k trials the judge could not rate, also counted as failures. Succ., Beh. (behaviour match) and Faith. (faithfulness) are estimates by the LLM judge of Section , **not human labels**. Judge–human agreement on strict success over 30 human-labelled items: 97% agreement, $\kappa = 0.93$ (1 reviewer(s)). Recall@5 and MRR measure whether the gold documents appear in the agent's search and read results (answerable questions; all cells are 0.98–0.99, so they cannot separate the configurations). Cit. P/R/F1 compare the cited document ids with the gold ids (nothing cited counts as 0); they are measured on document ids and say nothing about answer correctness. Red entries are not measured.

Benchmark	Configuration	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
HCMUT regulations	no verifier	121/23/3/3	2.0	*n/r*	*n/r*	*n/r*	10233/666	3.13	4	0.270
	verifier	109/35/3/3	2.0	*n/r*	*n/r*	*n/r*	12923/807	3.47	6	0.282
	verifier + reranker	109/37/1/3	2.0	*n/r*	*n/r*	*n/r*	12070/763	3.37	10	0.312

Table 4: Operating measurements and judge cost with the Bonsai 4B controller on the HCMUT regulations benchmark. Ans/Ref/Clar/Err are the numbers of answered, refused, clarified and execution-error trials; Err. is the execution-error rate. n/r (not reported): latency was recorded while several workers ran at once, so it is not reportable. Call = one controller model call; tok/s = generated tokens per second. Tokens are controller input/output per trial (trials with complete usage only). Judge cost is the usage cost reported by the API gateway over all retained attempts (1.064 USD for all configurations; not an invoice).

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
verifier vs no verifier	HCMUT regulations	50	-6.0 [-16.0,+4.0]	-5.3 [-14.7,+3.3]	+0.0 [+0.0,+0.0]	-11.6 [-22.0,-1.7]	+0.0 [-3.3,+2.7]
verifier + reranker vs verifier	HCMUT regulations	50	-3.3 [-13.3,+6.7]	-2.7 [-9.3,+3.3]	-1.1 [-3.3,+0.0]	+1.0 [-6.2,+7.7]	+0.0 [-2.7,+2.7]
verifier + reranker vs no verifier	HCMUT regulations	50	-9.3 [-18.7,+0.0]	-8.0 [-16.7,+0.0]	-1.1 [-3.3,+0.0]	-10.6 [-20.7,-0.5]	+0.0 [-4.0,+3.3]

Table 5: Differences between configurations on the HCMUT regulations benchmark (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over questions (seeds averaged per question; 5000 resamples). Succ. is strict success (LLM-judge lower bound), Beh. the behaviour match, Rec.@5 the Recall@5 of gold documents, Cit. F1 the citation F1 on document ids, Err. the execution-error rate. There is no multiple-comparison correction; an interval that includes 0 shows no detectable difference at this sample size.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
HCMUT regulations	*closed-book* (diagnostic)	50/50	2.0 [0–6]	0.0 [0–0]	2.2	–	–
	oracle (diagnostic)	45/45	82.2 [71–93]	82.2 [71–93]	100.0	1.000	1.000
	no verifier (reference)	135	–	69.6 [60–79]	88.1	0.844	0.807
	verifier (reference)	135	–	61.5 [50–73]	80.7	0.726	0.693
	verifier + reranker (reference)	135	–	59.3 [49–70]	79.3	0.733	0.704

Table 6: Two diagnostic baselines on the HCMUT regulations benchmark: one seed (42), the same 4B controller, sampler, domain prompt and constrained decoding, *no tools*, verifier off. They are not system configurations and are never pooled with them. *Closed-book*: the question only, all questions. *Oracle*: the full gold records are given up front in the format of a read result, answerable questions only (refusal questions have no gold context and are excluded). Succ. (all) is over every question the row ran; Succ. (answerable) is restricted to answerable questions (the like-for-like column), with a bootstrap interval over questions; both are LLM-judge lower bounds and errors count as failures. Answer rate = answered / all recorded trials (execution errors count as not answered) on answerable questions. Cit. P/R compare cited ids with gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the answerable-only results of the three configurations (seeds averaged). The oracle bounds what the 4B controller achieves with perfect retrieval; closed-book bounds what the model's own knowledge achieves.

Benchmark	Configuration	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
Labour law	no verifier	150	62.0 [52–72]$^{\dagger 7}$	91.6	87.3	0.989	0.952	0.886	0.830	0.841
	verifier	150	52.7 [43–63]$^{\dagger 6}$	75.7	86.3	0.989	0.952	0.759	0.719	0.728
	verifier + reranker	150	56.7 [47–67]$^{\dagger 4}$	76.7	85.6	0.993	0.959	0.726	0.678	0.691

Table 7: Results with the Bonsai 4B controller on the Labour law benchmark, three configurations (450 trials; random seeds 42, 123, 2026). *Succ.* is strict success (appropriate behaviour, correctness and completeness both full, every claim supported): a lower bound over all trials, with a 95% bootstrap interval over questions; execution errors count as failures, and $\dagger k$ marks k trials the judge could not rate, also counted as failures. Succ., Beh. (behaviour match) and Faith. (faithfulness) are estimates by the LLM judge of Section , **not human labels**. Judge–human agreement on strict success over 30 human-labelled items: 90% agreement, $\kappa = 0.80$ (1 reviewer(s)). Recall@5 and MRR measure whether the gold documents appear in the agent's search and read results (answerable questions; all cells are 0.99–0.99, so they cannot separate the configurations). Cit. P/R/F1 compare the cited document ids with the gold ids (nothing cited counts as 0); they are measured on document ids and say nothing about answer correctness. Red entries are not measured.

Benchmark	Configuration	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
Labour law	no verifier	132/14/4/0	0.0	*n/r*	*n/r*	*n/r*	9070/686	3.11	7	0.360
	verifier	111/31/4/4	2.7	*n/r*	*n/r*	*n/r*	11382/840	3.65	6	0.327
	verifier + reranker	108/37/1/4	2.7	*n/r*	*n/r*	*n/r*	11477/865	3.69	4	0.269

Table 8: Operating measurements and judge cost with the Bonsai 4B controller on the Labour law benchmark. Ans/Ref/Clar/Err are the numbers of answered, refused, clarified and execution-error trials; Err. is the execution-error rate. n/r (not reported): latency was recorded while several workers ran at once, so it is not reportable. Call = one controller model call; tok/s = generated tokens per second. Tokens are controller input/output per trial (trials with complete usage only). Judge cost is the usage cost reported by the API gateway over all retained attempts (1.103 USD for all configurations; not an invoice).

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
verifier vs no verifier	Labour law	50	-9.3 [-18.7,+0.0]	-12.7 [-20.7,-5.3]	+0.0 [+0.0,+0.0]	-11.2 [-19.8,-3.5]	+2.7 [+0.7,+5.3]
verifier + reranker vs verifier	Labour law	50	+4.0 [-6.0,+14.0]	-2.0 [-8.7,+5.3]	+0.4 [-2.2,+3.3]	-3.7 [-10.6,+3.2]	+0.0 [-2.7,+2.7]
verifier + reranker vs no verifier	Labour law	50	-5.3 [-15.3,+4.0]	-14.7 [-24.0,-6.0]	+0.4 [-2.2,+3.3]	-14.9 [-25.3,-5.7]	+2.7 [+0.0,+6.0]

Table 9: Differences between configurations on the Labour law benchmark (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over questions (seeds averaged per question; 5000 resamples). Succ. is strict success (LLM-judge lower bound), Beh. the behaviour match, Rec.@5 the Recall@5 of gold documents, Cit. F1 the citation F1 on document ids, Err. the execution-error rate. There is no multiple-comparison correction; an interval that includes 0 shows no detectable difference at this sample size.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	24.0 [12–36]	24.4 [13–38]	2.2	–	–
	oracle (diagnostic)	45/45	75.6 [62–87]	75.6 [62–87]	97.8	0.967	0.944
Labour law	no verifier (reference)	135	–	60.7 [50–71]	95.6	0.886	0.830
	verifier (reference)	135	–	51.1 [40–62]	80.7	0.759	0.719
	verifier + reranker (reference)	135	–	54.1 [43–64]	78.5	0.726	0.678

Table 10: Two diagnostic baselines on the Labour law benchmark: one seed (42), the same 4B controller, sampler, domain prompt and constrained decoding, *no tools*, verifier off. They are not system configurations and are never pooled with them. *Closed-book*: the question only, all questions. *Oracle*: the full gold records are given up front in the format of a read result, answerable questions only (refusal questions have no gold context and are excluded). Succ. (all) is over every question the row ran; Succ. (answerable) is restricted to answerable questions (the like-for-like column), with a bootstrap interval over questions; both are LLM-judge lower bounds and errors count as failures. Answer rate = answered / all recorded trials (execution errors count as not answered) on answerable questions. Cit. P/R compare cited ids with gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the answerable-only results of the three configurations (seeds averaged). The oracle bounds what the 4B controller achieves with perfect retrieval; closed-book bounds what the model's own knowledge achieves.

Benchmark	Configuration	Answer expected: A/R/C/Err	Refusal expected: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	no verifier	119/10/3/3	2/13/0/0	9.8	13.3	56.5	86.7	88.0
HCMUT regulations	verifier	109/20/3/3	0/15/0/0	17.4	0.0	42.9	100.0	82.7
	verifier + reranker	107/24/1/3	2/13/0/0	18.9	13.3	35.1	86.7	80.0

Table 11: Behaviour against what the benchmark expects on the HCMUT regulations benchmark (no judge). The rows "answer expected" and "refusal expected" give the numbers of answered / refused / clarified / error trials (A/R/C/Err). Over-refusal = refused or clarified although an answer exists; unsafe answer = answered although the question should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Benchmark	Configuration	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	no verifier	73.3 [62–84] (n=15)	67.8 [53–81] (n=30)	73.3 [47–100] (n=5)
HCMUT regulations	verifier	80.0 [64–93] (n=15)	52.2 [37–67] (n=30)	86.7 [60–100] (n=5)
	verifier + reranker	71.1 [56–84] (n=15)	53.3 [40–68] (n=30)	73.3 [33–100] (n=5)

Table 12: Strict success split by whether the question can be guessed on the HCMUT regulations benchmark (LLM-judge lower bound, errors count as failures). "Closed-book guessable": a strong model answered the question correctly *without* the corpus (itself an LLM estimate), so these questions measure the model's own knowledge as much as grounding. "Not guessable": partially correct or unsolved without the corpus. Refusal questions have no such split. Intervals are bootstrap intervals over questions; n = questions in the group.

Benchmark	Configuration	Answer expected: A/R/C/Err	Refusal expected: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	no verifier	129/2/4/0	3/12/0/0	4.4	20.0	85.7	80.0	94.0
Labour law	verifier	109/18/4/4	2/13/0/0	16.8	13.3	41.9	86.7	81.3
	verifier + reranker	106/24/1/4	2/13/0/0	19.1	13.3	35.1	86.7	79.3

Table 13: Behaviour against what the benchmark expects on the Labour law benchmark (no judge). The rows "answer expected" and "refusal expected" give the numbers of answered / refused / clarified / error trials (A/R/C/Err). Over-refusal = refused or clarified although an answer exists; unsafe answer = answered although the question should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Benchmark	Configuration	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	no verifier	52.8 [31–72] (n=12)	63.6 [51–76] (n=33)	73.3 [53–93] (n=5)
Labour law	verifier	38.9 [19–58] (n=12)	55.6 [42–68] (n=33)	66.7 [40–93] (n=5)
	verifier + reranker	52.8 [33–72] (n=12)	54.5 [41–68] (n=33)	80.0 [67–93] (n=5)

Table 14: Strict success split by whether the question can be guessed on the Labour law benchmark (LLM-judge lower bound, errors count as failures). "Closed-book guessable": a strong model answered the question correctly *without* the corpus (itself an LLM estimate), so these questions measure the model's own knowledge as much as grounding. "Not guessable": partially correct or unsolved without the corpus. Refusal questions have no such split. Intervals are bootstrap intervals over questions; n = questions in the group.